

**BOSTON BUOY 44013, MA, USA**WMO: **992420**Lat: **42.346N**Lon: **70.651W**Elev: **0**StdP: **14.70**Time Zone: **-5.00 (NAE)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.7       | 16.0 | 0.2                           | 5.5 | 15.4 | 2.9 | 6.4 | 17.8 | 39.6                  | 25.7 | 36.9 | 25.3 | 23.4                  | 310  | 0.640 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.2                    | 75.9            | 69.8 | 73.8 | 68.9 | 72.2 | 68.1 | 71.8                | 73.6 | 70.5 | 72.4 | 69.2 | 71.2 | 9.6                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.2                            | 115.2 | 72.3 | 69.8 | 109.5 | 71.4 | 68.5 | 104.8 | 70.4 | 35.5          | 73.7 | 34.4 | 72.5 | 33.4 | 71.4 |                   |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 34.5          | 31.2 | 26.5 | DB                         | 6.3 | 80.2               | 4.5 | 2.4                                                | 3.1 | 81.9       | 0.4  | 83.3       | -2.1 | 84.6       | -5.4 | 86.3 | (4) |
| (5)               |      |      | WB                         | 4.5 | 74.0               | 4.8 | 3.0                                                | 1.0 | 76.2       | -1.8 | 77.9       | -4.4 | 79.6       | -7.9 | 81.8 | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 49.7   | 31.5  | 32.2  | 36.9  | 44.4  | 52.6  | 61.1  | 67.6  | 67.5  | 62.9  | 54.5  | 46.0  | 38.3 |
|                                                                                  | DBStd       | 13.95  | 8.87  | 7.23  | 5.86  | 4.92  | 4.73  | 5.03  | 3.46  | 3.34  | 4.00  | 5.01  | 6.28  | 7.27 |
|                                                                                  | HDD50       | 2214   | 575   | 498   | 408   | 178   | 24    | 0     | 0     | 0     | 0     | 18    | 146   | 367  |
|                                                                                  | HDD65       | 5806   | 1040  | 918   | 872   | 617   | 386   | 134   | 14    | 14    | 86    | 327   | 569   | 829  |
|                                                                                  | CDD50       | 2112   | 0     | 0     | 1     | 11    | 104   | 335   | 545   | 541   | 388   | 157   | 27    | 3    |
|                                                                                  | CDD65       | 229    | 0     | 0     | 0     | 0     | 1     | 19    | 94    | 90    | 24    | 1     | 0     | 0    |
|                                                                                  | CDH74       | 154    | 0     | 0     | 0     | 0     | 1     | 10    | 81    | 53    | 9     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.4   | 17.9  | 16.9  | 15.4  | 12.7  | 11.0  | 9.7   | 9.0   | 9.4   | 11.2  | 14.3  | 15.9  | 17.6 |
| Precipitation                                                                    | PrecAvg     | 50.8   | 4.2   | 3.7   | 5.4   | 4.4   | 4.0   | 3.7   | 3.5   | 4.1   | 4.1   | 4.6   | 4.1   | 4.7  |
|                                                                                  | PrecMax     | 60.8   | 7.0   | 8.0   | 16.2  | 9.3   | 12.3  | 12.6  | 8.5   | 6.5   | 8.7   | 12.7  | 7.4   | 8.2  |
|                                                                                  | PrecMin     | 39.2   | 1.5   | 2.0   | 0.4   | 1.1   | 1.2   | 0.1   | 1.6   | 0.9   | 1.0   | 0.6   | 0.9   | 1.5  |
|                                                                                  | PrecStd     | 6.5    | 1.5   | 1.7   | 3.4   | 2.2   | 2.4   | 3.1   | 1.7   | 1.8   | 2.3   | 2.9   | 1.9   | 2.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.8  | 50.4  | 54.4  | 61.3  | 69.2  | 75.4  | 78.7  | 78.2  | 75.3  | 68.4  | 62.0  | 57.4 |
|                                                                                  |             | MCWB   | 51.0  | 47.4  | 47.5  | 54.7  | 62.3  | 67.3  | 71.5  | 71.0  | 70.9  | 63.4  | 59.2  | 54.2 |
|                                                                                  | 2%          | DB     | 48.6  | 46.0  | 49.4  | 56.6  | 64.5  | 72.0  | 76.3  | 75.4  | 72.2  | 64.9  | 58.4  | 52.9 |
|                                                                                  |             | MCWB   | 46.3  | 43.3  | 45.2  | 51.5  | 59.5  | 65.6  | 69.9  | 69.8  | 68.5  | 61.0  | 55.1  | 50.4 |
|                                                                                  | 5%          | DB     | 45.6  | 43.1  | 46.3  | 53.1  | 61.6  | 69.6  | 74.1  | 73.4  | 70.1  | 62.6  | 56.1  | 50.5 |
|                                                                                  |             | MCWB   | 43.0  | 40.3  | 42.8  | 49.0  | 57.2  | 64.7  | 69.1  | 69.1  | 67.1  | 59.4  | 52.8  | 48.6 |
|                                                                                  | 10%         | DB     | 43.0  | 41.2  | 43.8  | 50.6  | 59.0  | 67.6  | 72.4  | 72.0  | 68.3  | 60.9  | 54.4  | 48.2 |
|                                                                                  |             | MCWB   | 40.4  | 38.5  | 40.8  | 46.8  | 55.6  | 63.7  | 68.1  | 68.4  | 65.6  | 57.9  | 51.4  | 46.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 50.3  | 47.9  | 49.3  | 56.9  | 64.2  | 69.5  | 74.5  | 73.4  | 73.4  | 65.4  | 60.2  | 54.8 |
|                                                                                  |             | MCDB   | 50.9  | 48.9  | 50.5  | 60.7  | 67.3  | 72.8  | 76.4  | 74.8  | 74.1  | 67.5  | 61.3  | 55.8 |
|                                                                                  | 2%          | WB     | 46.0  | 43.4  | 45.7  | 53.4  | 61.0  | 67.5  | 71.6  | 71.7  | 70.5  | 62.7  | 56.2  | 51.2 |
|                                                                                  |             | MCDB   | 47.3  | 44.7  | 47.8  | 56.6  | 63.1  | 70.3  | 73.8  | 73.5  | 71.6  | 64.1  | 57.5  | 52.5 |
|                                                                                  | 5%          | WB     | 42.6  | 40.7  | 43.3  | 50.3  | 58.6  | 65.9  | 70.3  | 70.5  | 68.4  | 60.8  | 53.8  | 49.1 |
|                                                                                  |             | MCDB   | 44.1  | 42.1  | 45.2  | 52.8  | 59.9  | 68.3  | 72.5  | 72.3  | 69.7  | 62.2  | 55.3  | 50.2 |
|                                                                                  | 10%         | WB     | 39.7  | 38.4  | 41.3  | 48.0  | 56.2  | 64.4  | 69.1  | 69.3  | 66.7  | 59.1  | 52.0  | 46.5 |
|                                                                                  |             | MCDB   | 41.9  | 40.2  | 43.4  | 50.3  | 58.0  | 66.7  | 71.4  | 71.1  | 68.2  | 60.5  | 53.9  | 47.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.7   | 9.1   | 7.9   | 7.0   | 7.3   | 7.1   | 6.9   | 6.2   | 6.0   | 6.9   | 7.3   | 8.7  |
|                                                                                  |             | MCDBR  | 11.7  | 10.8  | 10.8  | 11.7  | 12.1  | 10.6  | 9.5   | 8.5   | 8.4   | 8.7   | 8.9   | 10.3 |
|                                                                                  |             | MCWBR  | 11.9  | 10.7  | 8.4   | 8.4   | 8.4   | 6.5   | 5.5   | 5.6   | 6.4   | 7.0   | 9.0   | 10.5 |
|                                                                                  | 5% WB       | MCDBR  | 11.0  | 10.0  | 9.9   | 11.7  | 11.1  | 9.3   | 8.3   | 7.5   | 7.6   | 7.4   | 8.5   | 10.1 |
|                                                                                  |             | MCWBR  | 12.1  | 10.7  | 9.0   | 9.3   | 8.9   | 6.6   | 5.8   | 5.9   | 6.6   | 7.2   | 9.5   | 11.3 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.298  | 0.305 | 0.325 | 0.369 | 0.405 | 0.455 | 0.475 | 0.435 | 0.378 | 0.338 | 0.313 | 0.299 |      |
|                                                                                  | taud        | 2.525  | 2.515 | 2.506 | 2.407 | 2.349 | 2.245 | 2.196 | 2.296 | 2.439 | 2.549 | 2.577 | 2.575 |      |
|                                                                                  | Ebn at Noon | 272    | 288   | 293   | 285   | 276   | 262   | 255   | 262   | 271   | 272   | 264   | 262   |      |
|                                                                                  | Edh at Noon | 22     | 26    | 30    | 36    | 39    | 43    | 45    | 39    | 32    | 25    | 21    | 20    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 543    | 821   | 1144  | 1481  | 1663  | 1792  | 1884  | 1694  | 1357  | 906   | 605   | 451   |      |
|                                                                                  | RadStd      | 37     | 73    | 80    | 160   | 177   | 164   | 115   | 69    | 89    | 76    | 46    | 50    |      |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (68 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +167  | +76   |

Nomenclature: See separate page